

Backend Study Roadmap

A Step-by-Step Guide for Manoj

Step 1: Strong JavaScript Foundation

- Understand variables, functions, arrow functions.
- Learn async/await and Promises.
- Practice array methods (map, filter, reduce).
- Understand objects, destructuring, and modules.
- Goal: Be comfortable writing JavaScript without confusion.

Step 2: Node.js Basics

- Understand what Node.js is and how a server works.
- Learn about the event loop.
- Create a basic HTTP server.
- Understand request and response handling.

Step 3: Express.js Framework

- Learn routing (GET, POST, PUT, DELETE).
- Understand middleware.
- Handle JSON data.
- Build small APIs like a To-Do API.

Step 4: Database (MongoDB)

- Understand collections and documents.
- Learn CRUD operations.

- Use Mongoose for schemas and models.
- Connect backend to database.

Step 5: Authentication System

- Learn bcrypt for password hashing.
- Learn JWT for authentication tokens.
- Create signup and login APIs.
- Protect routes using middleware.

Step 6: Advanced Backend Topics

- Google login using Passport.
- File upload using Multer.
- Email sending using Nodemailer.
- Payment integration basics.
- Error handling and security best practices.

Step 7: Real Projects

- Build a full authentication system.
- Create a Blog API.
- Develop a mini e-commerce backend.
- Build a chat backend system.
- Implement a file upload system.

Weekly Study Plan Example

- Week 1: JavaScript revision.
- Week 2: Node + Express basics.
- Week 3: MongoDB + CRUD.
- Week 4: Authentication (bcrypt + JWT).

- Week 5: Build a full project.

Golden Study Rules

- Do not copy-paste code blindly.
- Write code yourself.
- Break and debug your code.
- Build small projects regularly.
- Consistency is more important than speed.